The Recipe, Not the Data: An Audited Ledger of Physics Identifiability in a Structured World Model

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Abstract

We present an end-to-end audited study of physical-parameter identifiability in a small structured world model, across a toy 2D environment and a MuJoCo contact-physics environment. Our starting point is a retraction: the project's previous conclusion — that a friction parameter was "not identifiable" under random exploration — was wrong, caused by an uncommitted evaluation oracle whose MSE objective was destroyed by heavy-tailed collision outliers, compounded by a simulator bug that silently voided the physics signal. Re-verifying every claim produced a three-step thesis, each step measured under episode-held-out evaluation: (1) *the information exists* — a robust closed-form estimator recovers friction at R^2=0.87 from the same noisy data where a GRU belief encoder scores −0.49; (2) *features unlock learning* — handing the estimator's sufficient statistics to the same GRU flips all three parameters positive (gravity +0.43); (3) *collection amplifies* — an information-seeking policy improves belief quality +39% over random data with the model held fixed. The thesis transfers to MuJoCo contact physics (friction +0.28, mass +0.11 from a zero baseline), where analysis also yields a structural identifiability result (only the μg product enters box dynamics) and a cautionary finding: deterministic information-seeking policies require *anti-fragility to unobservables.* All claims map to committed scripts, seeded runs, and regression tests.

1 Introduction

Negative identifiability results in world-model research are commonly attributed to the *data regime.* We document a case where that conclusion was published in a project's model card and was wrong three ways at once: the supporting oracle was never committed (unreproducible), its objective was statistically fragile (MSE under heavy-tailed outliers), and the environment itself had a bug that removed the signal being measured. Rather than merely fixing the record, we use the retraction as a method: every subsequent claim is (i) reproduced by a committed script, (ii) evaluated with episode-grouped splits (row-level splits leak episode labels through overlapping windows — we measure the leak at $R^2 \approx 0.3$ on pure noise), and (iii) guarded by a regression test.

2 Setup

Environments. (a) *CruelGridworld*: three bodies on a bounded plane, nonlinear inter-object attraction (G/d^2, active $1<d<10$), per-episode gravity/friction randomization, process noise σ=0.05, 8 discrete force actions, 6-D position-only observations. (b) *MujocoPointMass*: an actuated ball and two passive boxes on a bounded MuJoCo plane; episode latents are sliding friction, box masses and gravity; same discrete action interface plus a no-op. **Belief model.** A GRU over K-step same-episode windows (RMA/VariBAD lineage [1, 2]) with a heteroscedastic Gaussian head and an optional half-window InfoNCE episode-contrastive term. **Protocol.** Episode-grouped splits throughout; supervised held-out R^2 per parameter; fully seeded training with a bit-identical canary test.

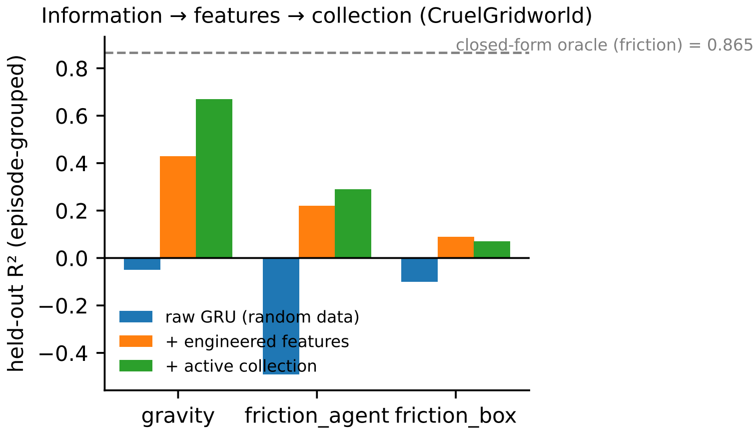

Figure 1: The three-step thesis on CruelGridworld: same GRU, same evaluation; only features and data collection change.

3 A retraction as a methodology lesson

A committed median-of-ratios estimator on position-only observations — per-step decay ratios $\langle v', v{+}\alpha u\rangle/\|v{+}\alpha u\|^2$, gated by excitation, boundary-bounce and interaction filters, aggregated by the median — recovers friction at $R^2{=}0.865$ (MAE 0.006) on the same noisy random-policy data previously declared uninformative. The original oracle failed because collisions create heavy-tailed ratio outliers: a least-squares fit of the identical quantity scores $R^2{=}-35$. Separately, a containment bug let passive boxes exit the arena, silently removing the interaction signal. *Negative identifiability claims inherit every fragility of their oracle and their simulator.*

4 The three-step thesis

Step 1 — the information exists. The closed-form estimator sets the reference (friction 0.865); the GRU on raw windows scores -0.49 on the same data while its training loss decreases — episode memorization, not learning. **Step 2 — features unlock learning.** Handing the GRU the estimator's sufficient statistics (gated decay ratio and its running median, excitation, distances, attractor-projected box accelerations, $1/d^2$ regressors, aligned actions) flips all three parameters positive (Fig. 1). Two measured failure modes: a single misaligned action index nullifies every ratio; and an *evidence-length bound* — the target's std (0.025) lies below the optimal estimator's window-level error at 18 steps (0.043), so no learner can score positive there. **Step 3 — collection amplifies.** An information-seeking policy raises belief quality +39% with the model fixed (gravity 0.67). Its first version dashed along a fixed line and repeatedly struck an unobservable obstacle, poisoning the episode median (Fig. 2); a golden-angle rosette restores robustness — *anti-fragility to unobservables.*

5 Transfer to MuJoCo contact physics

Random policy + generic statistics: $R^2{\approx}0$ on all parameters. Analysis explains why: the actuated agent rides slide joints (normal force $\approx$0; it slides nearly frictionless), so only the boxes feel friction; Coulomb friction decelerates *linearly* (ratio features are the wrong model class); and gravity enters box dynamics only through μg — *structurally unidentifiable alone*, an exclusion made with a measurement rather than by assertion. A COAST/PUSH policy (no-ops manufacture free sliding; box-directed pushes manufacture contacts) plus aligned-impulse features (head-on contacts only; glancing hits confound geometry with mass at -0.27) yields learned beliefs of friction +0.28 and mass +0.11 at 1,048 episodes (Fig. 3).

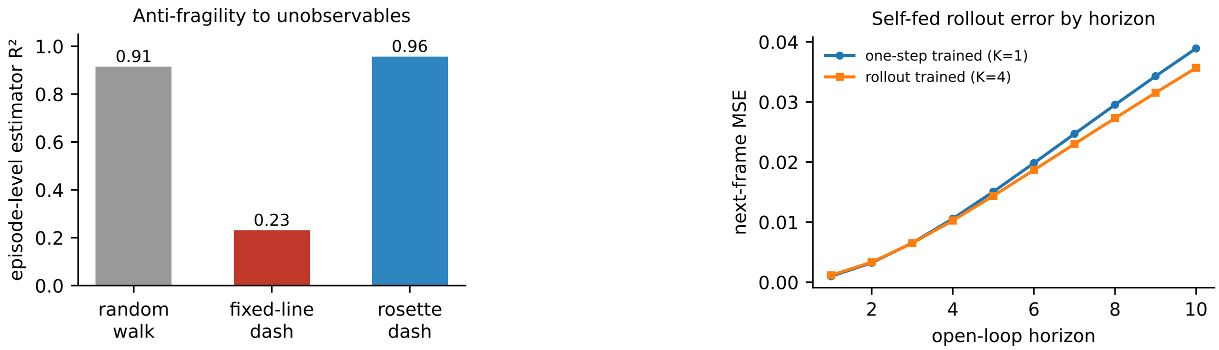

Figure 2: Left: deterministic information-seeking must be robust to unobservables. Right: rollout-trained world models trade one-step accuracy for slower open-loop degradation.

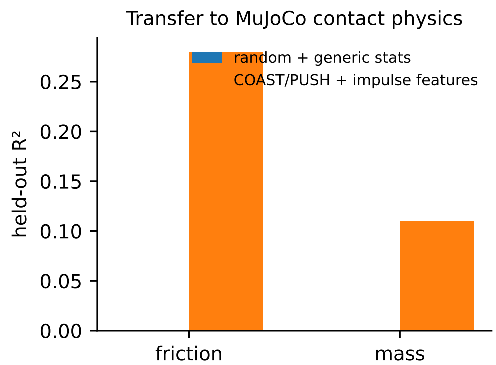

Figure 3: The thesis transfers to real contact physics.

6 Related work

Belief-style system identification: RMA [1], VariBAD [2]. Active system identification: ASID [3] — our policies replace the Fisher objective with measured information-rate proxies. Self-predictive stability: Ni et al. [4]; our world model uses VICReg-style regularization [5] plus prediction grounding. Our contribution is an audited, fully reproducible account connecting these pieces, with retraction, failure ledger, and structural identifiability analysis.

7 Limitations

Two synthetic environments; no visual observations; policies are designed, not learned; the world model sits 2.3× above its linear information ceiling on MuJoCo; mass identification remains weak (+0.11).

References

[1] A. Kumar, Z. Fu, D. Pathak, J. Malik. RMA: Rapid Motor Adaptation for Legged Robots. *RSS*, 2021.

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[3] M. Memmel et al. ASID: Active Exploration for System Identification. *ICLR*, 2024.

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[5] A. Bardes, J. Ponce, Y. LeCun. VICReg: Variance-Invariance-Covariance Regularization. *ICLR*, 2022.